Side A — core likelihoods and model families

Lecture-grounded formulas and facts only.

FORMULAS + CONCEPTS

FOUNDATIONS

1. Density, KL, MLE

$$\mathrm{KL}(P_{\text{data}}||P_\theta) = \mathbb{E}_{P_{\text{data}}}[\log P_{\text{data}}(x) - \log p_\theta(x)]$$

$$\theta^* = \arg\max_\theta \frac{1}{N}\sum_{i=1}^{N} \log p_\theta(x_i)$$

$$p_\theta(x) = \int p_\theta(x,z)\,dz \quad \text{or} \quad \sum_z p_\theta(x,z)$$

- Why MLE: minimizing KL(pdata||pθ) differs only by fixed data entropy, so maximize expected log pθ(x).
- Latent model: specify easy joint pθ(x,z); likelihood needs summing/integrating all latent explanations z.
- Hardness: posterior pθ(z|x)=pθ(x,z)/pθ(x) needs the same marginal; exact inference is rare.
- Exam intuition: if z explains x, learning must reward both good reconstruction and plausible latent mass.

EM PPCA

2. EM and PPCA

$$\mathcal{L}(q,\theta) = \mathbb{E}_{q(z)}[\log p_\theta(x,z)] + H(q)$$

E: $q(z) \leftarrow p_{\theta_{\text{old}}}(z \mid x)$ M: $\theta \leftarrow \arg\max_\theta \mathbb{E}_{q(z)}[\log p_\theta(x,z)]$

$$z \sim \mathcal{N}(0, I), \quad x \mid z \sim \mathcal{N}(Wz+\mu, \sigma^2 I), \quad x \sim \mathcal{N}(\mu, WW^\top + \sigma^2 I)$$

- EM reasoning: E-step makes the bound tight at θold; M-step improves θ under that posterior weighting.
- ELBO view: q handles uncertainty over z; entropy H(q) prevents choosing one latent explanation too cheaply.
- PPCA intuition: Wz creates correlated low-dimensional structure; σ²I is leftover isotropic noise.
- Limitation: EM/PPCA are clean because Gaussian/linear; nonlinear decoders break closed-form updates.

VAE

3. VAE / amortized VI

$$\log p_\theta(x) = \log \int p_\theta(x,z)\,dz \geq \mathbb{E}_{q_\phi(z \mid x)}[\log p_\theta(x,z) - \log q_\phi(z \mid x)]$$

$$\mathcal{L}(\theta,\phi;x) = \mathbb{E}_{q_\phi(z \mid x)}[\log p_\theta(x \mid z)] - \mathrm{KL}(q_\phi(z \mid x)\|p(z))$$

$$\log p_\theta(x) = \mathcal{L}(\theta,\phi;x) + \mathrm{KL}(q_\phi(z \mid x)\|p_\theta(z \mid x))$$

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I)$$

- Cannot directly optimize log pθ(x) because the marginal integral and true posterior are intractable for nonlinear decoders.
- Lower-bound proof: insert q(z|x), rewrite as expectation, apply Jensen; gap is KL(qφ(z|x)||pθ(z|x)) ≥ 0.
- Two terms: reconstruction/data-fit makes x likely given z; KL-to-prior regularizes latent space for sampling.
- Amortized VI: encoder network predicts qφ parameters per x instead of optimizing a separate q for every datapoint.
- Reparameterization: z=μ+σ⊙ε makes stochastic sampling differentiable w.r.t. μ,σ,φ.

ENERGY

4. Energy-based models

$$p_\theta(x) = \frac{\exp(-E_\theta(x))}{Z_\theta}, \qquad Z_\theta = \int \exp(-E_\theta(x))\,dx$$

$$\log p_\theta(x) = -E_\theta(x) - \log Z_\theta$$

$$\nabla_\theta \log p_\theta(x) = -\nabla_\theta E_\theta(x) + \mathbb{E}_{x' \sim p_\theta}[\nabla_\theta E_\theta(x')]$$

- PDF proof: exp(-Eθ(x)) > 0; Zθ is the integral of exp(-Eθ), so dividing by Zθ makes total mass 1.
- Flexibility: Eθ can be almost any neural scoring function; no need for normalized directed likelihood architecture.
- Gradient derivation: ∇logp=-∇E-∇logZ and ∇logZ = -Epθ[∇E], giving data phase minus model phase.
- Qualitative terms: lower data energies; raise energies of samples the model currently believes too much.
- MCMC need: model expectation over pθ is intractable; simulate chain with stationary distribution pθ. Higher energy = lower probability.

FLOW

5. Normalising flows

$$x = g_\theta(z), \quad z = f_\theta(x) = g_\theta^{-1}(x)$$

$$p_X(x) = p_Z(f_\theta(x))\left|\det J_{f_\theta}(x)\right|$$

$$\log p_X(x) = \log p_Z(f_\theta(x)) + \log\left|\det J_{f_\theta}(x)\right|$$

- Validity: invertible map preserves probability mass; determinant corrects local volume change.
- Exact likelihood: evaluate base density at inverse fθ(x) plus log|detJ|, no variational lower bound needed.
- Trade-off: expressive transforms are expensive; tractable triangular/coupling designs constrain architecture.
- Exam trap: forgetting the log determinant means ignoring whether the transform stretches or compresses space.

SCORE

6. Score-based models

$$s(x) = \nabla_x \log p(x)$$

$$p(x) = \frac{\exp(-E(x))}{Z} \Rightarrow \nabla_x \log p(x) = -\nabla_x E(x)$$

$$\min_\theta \mathbb{E}_{p_{\text{data}}}\left\|s_\theta(x) - \nabla_x \log p_{\text{data}}(x)\right\|^2$$

$$x_{t+1} = x_t + \eta s_\theta(x_t) + \sqrt{2\eta}\,\epsilon_t$$

- Direct density modelling needs nonnegative normalized pθ; normalization is the hard part.
- Score benefit: ∇x log p(x) ignores constants like Z and gives direction toward higher density.
- Score matching terms: norm penalizes large/uncontrolled scores; divergence term prevents trivial zero field and matches data geometry.
- Finite data issue: empirical distribution is spikes/deltas, so its score is useless between samples.
- Fix: add noise and learn noise-conditional/denoising scores; Langevin adds noise to sample the distribution, not just ascend one mode.

Side B — diffusion, flow matching, guidance, evaluation

Use as a memory sheet: equation, object meaning, limitation.

FORMULAS + CONCEPTS

DIFFUSION

7. Diffusion models

$$q(x_t \mid x_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\, x_{t-1}, \beta_t I)$$

$$q(x_t \mid x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\, x_0, (1-\bar{\alpha}_t) I)$$

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon$$

$$\mathbb{E}_{x_0,\varepsilon,t}\left\|\varepsilon - \varepsilon_\theta(x_t, t)\right\|^2$$

- Forward noising is fixed and tractable: every xt is a noisy version of x0 with known Gaussian form.
- Learning target εθ: if the model predicts added noise, it also knows how to denoise one reverse step.
- Sampling intuition: start from pure noise and repeatedly remove predicted noise while preserving uncertainty.
- Relation to score: denoising/noise prediction estimates the direction toward higher noisy-data density.
- Reason for stochasticity: avoids deterministic collapse and represents the whole reverse distribution.

FLOW-MATCHING

8. Continuous flows / Flow Matching

$$\frac{dx_t}{dt} = v_\theta(t, x_t)$$

$$\frac{d\log p_t(x_t)}{dt} = -\operatorname{div} v_\theta(t, x_t)$$

$$\min_\theta \mathbb{E}_{t,x_t}\left\|v_\theta(t, x_t) - u_t(x_t)\right\|^2$$

- CNF intuition: particles move continuously; density decreases where velocity field expands volume.
- Divergence is the continuous analogue of log determinant in normalizing flows.
- Flow matching target: choose an interpolation path, then train vθ to match the true velocity u_t at sampled points.
- Reason it helps: supervised velocity regression avoids likelihood ODE simulation inside each training objective.

GUIDANCE

9. Guidance / conditioning

$$\nabla_x \log p(x \mid y) = \nabla_x \log p(x) + \nabla_x \log p(y \mid x)$$

$$\varepsilon_{guided} = \varepsilon_{uncond} + w(\varepsilon_{cond} - \varepsilon_{uncond})$$

- Classifier guidance follows Bayes: conditional score = prior/sample score + gradient making y more likely.
- Classifier-free guidance approximates the same push by subtracting unconditional from conditional predictions.
- Guidance scale w is a trade-off: stronger condition adherence can reduce diversity or realism.
- For scientific generation, constraints and physical validity are part of the model objective, not decoration.

EVALUATION

10. Evaluation of generative models

$$\mathrm{IS} = \exp\left(\mathbb{E}_x \mathrm{KL}(p(y \mid x) \| p(y))\right)$$

$$\mathrm{FID} = \|\mu_r - \mu_g\|^2 + \operatorname{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\right)$$

$$\mathrm{MMD}^2(P, Q) = \mathbb{E}k(x, x') + \mathbb{E}k(y, y') - 2\mathbb{E}k(x, y)$$

- No single metric proves generative quality: likelihood, fidelity, diversity, and memorization can disagree.
- FID intuition: compare Gaussian moments of real/generated features; lower means closer feature distribution.
- Precision vs recall: precision = samples look real/on manifold; recall = model covers data modes.
- IS rewards confident diverse classifier labels but can miss memorization and domain mismatch.
- Always state metric failure mode: feature extractor bias, finite-sample noise, mode dropping, copying.

LIFE SCIENCES

11. Generative models for life sciences

- Scientific generative modelling is constrained generation: samples must be valid under chemistry/biology rules.
- Novelty alone is not enough; check usefulness, stability, toxicity, synthesizability, and uncertainty.
- Conditional generation/property optimization creates a quality-diversity trade-off like guidance.
- Main risk: model exploits proxy metrics or memorizes known structures instead of discovering useful candidates.